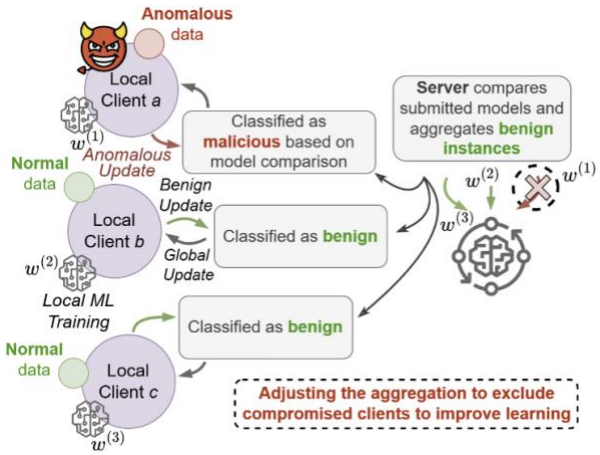
Problem: Anomaly Detection for Malicious Updates in Federated Learning

Federated Learning (FL) is a privacy-preserving AI paradigm where clients train a shared model locally and send only mathematical updates (weights) to a central server for aggregation. However, this privacy creates a security vulnerability: the server is *blind* to raw data and must implicitly trust every update it receives. Adversarial attacks exploit this trust (a top AI risk under OWASP LLM04). A malicious client can easily join the network and send "poisoned" updates, trained on corrupt data, to sabotage the global model or plant backdoors.



Your mission is to solve this problem by building an Anomaly Detector for these malicious updates. You will build an FL system and simulate attacks like "label flipping" (flipping local training labels to steer the global model off course). The core hypothesis is that a poisoned update is a statistical outlier compared to honest updates. Your "hack" will be to implement this detection logic inside the server's aggregation function (see figure). Stateless defenses check one round at a time by computing a center (e.g., median) and measuring L2 distance, cosine similarity, or pairwise distances (Krum) to flag updates "too far" away. However, recent FL security research show that single-round checks fail against *stealthy multi-round ("drip") attacks*, falsely ban honest clients with diverse data (*Non-IID*), and scale poorly at $O(n^2d)$, whereas stateful defenses like **PID-MADE** (tracking Proportional distance, Integral drift, and Derivative shifts) or Two-Sided Norm Screening (**TNBS**) can do a better job.

Thus, your **RESEARCH TASK** is to investigate what defense better detects malicious client updates (under IID and Non-IID data) and excludes them from aggregation. You may implement your own defense or use an existing one. Below is what is **GIVEN TO YOU** to get started on the task, but you are not limited to use any other dataset and/or papers:

- **Fundamental paper on FL:** FedAvg [https://arxiv.org/abs/1602.05629];
- **For the FL experiments, you are recommended to use Flower** [https://flower.ai/];
- **For the ML part, you are recommended to use PyTorch, which is integrated with Flower** [https://flower.ai/docs/framework/tutorial-quickstart-pytorch.html];
- **For the dataset, use FEMNIST** [https://huggingface.co/datasets/flwrlabs/femnist] or Flower Datasets [https://flower.ai/docs/datasets/];
- **For the attack modeling, start with "label flipping"** (randomly flipping labels on some clients), or explore model poisoning [https://arxiv.org/abs/1911.11815] and backdoors [https://arxiv.org/abs/1807.00459];
- **Examples of defenses you can implement for evaluation:** Krum/Multi-Krum [https://arxiv.org/abs/1703.02757], Median/Trimmed Mean [https://arxiv.org/abs/1803.01498] (*hint:* many defenses are already implemented in Flower, search and read the documentation!)

You need to report the results on evaluation (including round-by-round learning trajectories), the complexity of the implemented defenses (e.g., $O(n^2d)$ vs. $O(nd)$), their effectiveness (e.g., accuracy of malicious clients detection vs. false exclusions on Non-IID data), and present visualizations.